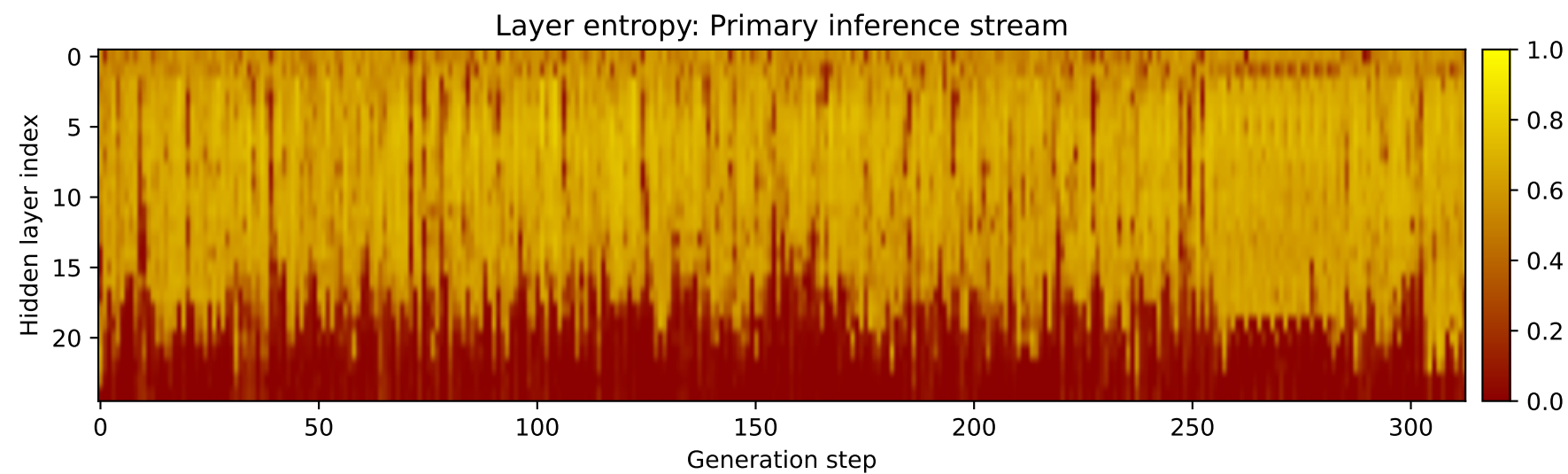
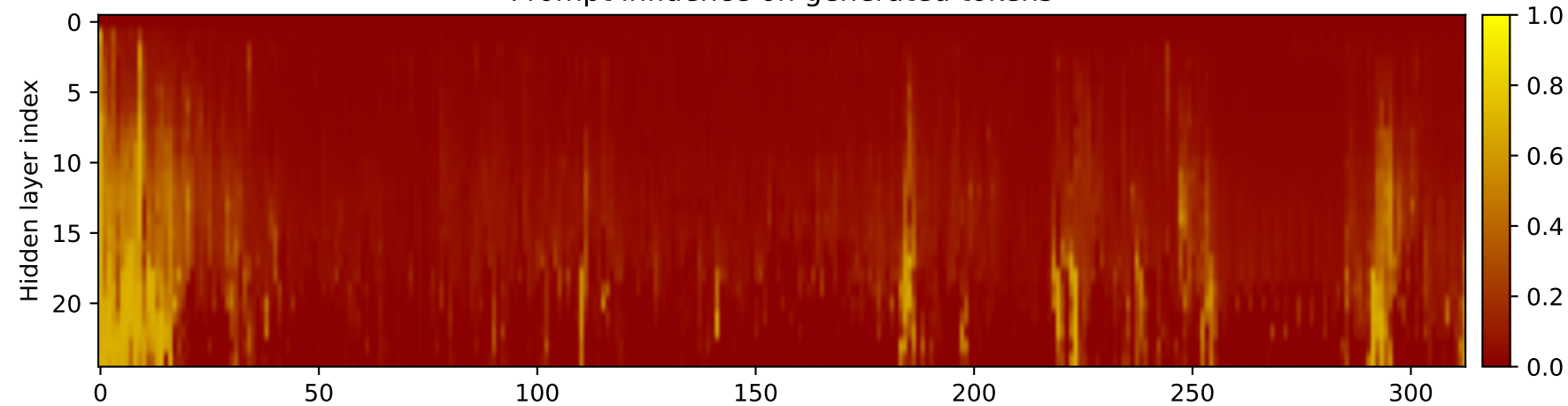
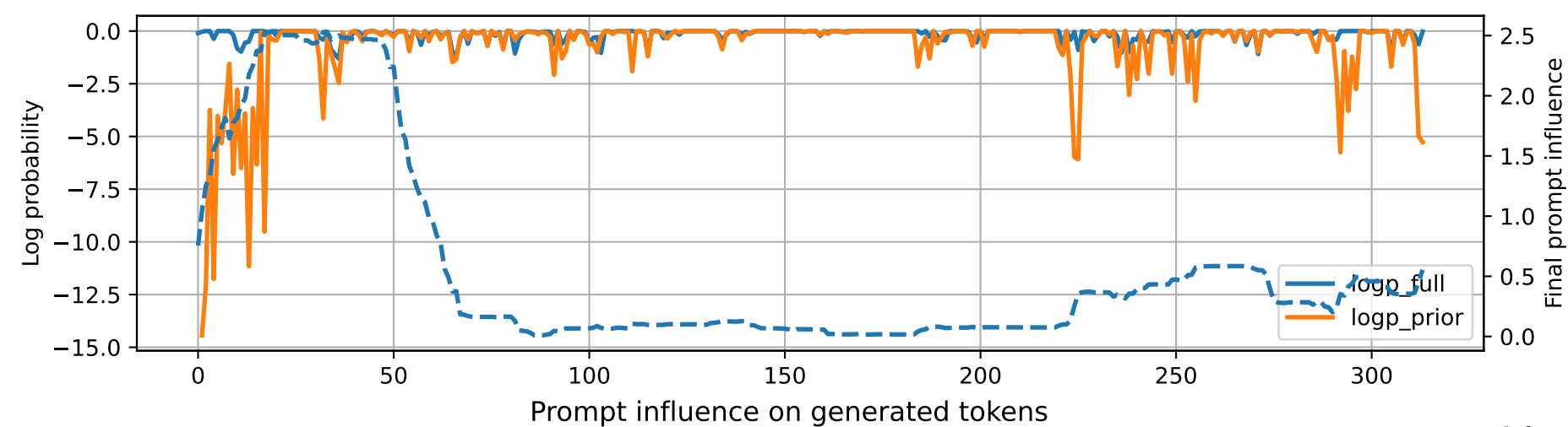


Prompt 1



PROMPT:
Task ID: PYC080 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC080 Your code must contain this required symbol:
def top_k_frequent Implement this task:
define function top_k_frequent(items,k).. Start your answer with
the check-token line, then provide the
Python implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC080 # Define function top_k_frequent(items, k)
def top_k_frequent(items, k): # Check if
the input arguments are valid if not isinstance(items, list)
or not isinstance(k, int) or k < 1:
raise ValueError("Both arguments must be a list and an integer
greater than 0.") # Create a dictionary to
store the frequency of each item frequency_dict = {} #
Iterate through the items in the list for
item in items: # Check if the item is already in the
dictionary if item in frequency_dict:
If it is, increment its frequency by 1
frequency_dict[item] += 1 else: # If
it is not, add it to the dictionary with a frequency of 1
frequency_dict[item] = 1 # Sort the
dictionary by frequency in descending order
sorted_frequency_dict = dict(sorted(frequency_dict.items(),
key=lambda x: x[1], reverse=True)) # Return the top k
frequent items return sorted_frequency_dict[:k]
Return the first k items with the highest frequency # Test the
function items = [1, 2, 3, 4, 5, 6, 7, 8, 9,
10] k = 3 print(top_k_frequent(items, k)) # Output: [4, 5, 6]